

Volume with Whole Number Edges Flagship

Lesson 10-1-flagship

Name: _____ Date: _____ Class: _____

CLASS LEGACY MISSION

Seal the Time Capsule

You are the lead builder for the school's 50-year time capsule project. Every memory must fit inside rectangular containers, and the principal needs to know exactly how much each one holds before they are buried. Master volume and the capsule gets sealed for the future.

Key Vocabulary Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box $3 \times 2 \times 4$ holds 24 unit cubes, so $V = 24$ cubic units

Volume

How much space is inside a solid shape.

A cereal box or shoe box — it has length, width, and height

Rectangular prism

A solid box shape with six flat rectangle sides.

A tiny cube that is $1 \text{ in} \times 1 \text{ in} \times 1 \text{ in} = 1 \text{ in}^3$ (one cubic inch)

Cubic units

The units used to measure space inside, like cubic inches.

$V = l \times w \times h$: for a box $5 \times 3 \times 2$, volume = 30 cubic units

Length, width, height

How long, how wide, and how tall a box is.

A cross-shaped pattern of 6 rectangles folds into a rectangular box

Net

A flat shape that folds up into a solid.

A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

Edge

The line where two flat sides of a solid meet.

Key Ideas & Notes

- Your team is building time capsule containers to store class memories.
- Each container is a rectangular prism, and you need to figure out how much each one can hold.
- That means calculating the volume using whole number dimensions!
- Calculate the volume of each time capsule container. Use $V = l \times w \times h$.

Think About It

- What three measurements do you need to find the volume of a box?
- What does volume tell us about a container?
- How is volume different from the area of one face?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

You are sealing time capsule containers and must know exactly how much each holds. Why do you need three edge measurements (length, width, and height) to find the volume of a container?

Level 1 support

Start here: Volume measures space in three directions, so you need length, width, and height.

I need three measurements because volume fills space in ____ directions.

Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give one ____, not the whole volume.

El largo y el ancho solos solo dan una ____, no todo el volumen.

Word bank: volume length width height rectangular prism edge

Listen for: Students explain volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only describe one flat face.

Level 2

If two containers must have the same volume but different shapes, how could you design them? Justify with an example.

- Two containers ____ have the same volume but different shapes because ____.
- For example, ____ × ____ × ____ and ____ × ____ × ____ both equal ____.

EXPLORE

In the Fold Shop you fold a flat net into a closed box. How does folding the net help you understand what the volume $V = l \times w \times h$ actually measures?

Level 1 support

Start here: The folded net encloses space, and that enclosed space is the volume you count in cubic units.

When the net folds up, the ____ faces enclose a ____ that I can fill with unit cubes.

Cuando la red se dobla, las ____ caras encierran un ____ que puedo llenar con cubos unitarios.

The volume is the number of ____ that fit inside, found by ____ $\times$ ____ $\times$ ____.

El volumen es el número de ____ que caben dentro, hallado por ____ $\times$ ____ $\times$ ____.

Word bank: **volume** **cubic units** **rectangular prism** **net** **edge**

Listen for: Students connect the folded net to the 3D space inside, and explain volume counts the unit cubes that fill it, computed as $l \times w \times h$.

Level 2

Container C ($8 \times 3 \times 4 = 96 \text{ in}^3$) holds more than Container B ($5 \times 5 \times 2 = 50 \text{ in}^3$).

Explain how the dimensions, not just one big edge, decide the volume.

- Container C holds more because all three dimensions multiply to ____.
- A single large edge does not guarantee a larger volume because ____.

CONNECT

At the Loading Dock, a large moving box is $24 \times 18 \times 18$ in and a medium box is $18 \times 14 \times 12$ in. Talk through how to find how many more cubic inches the large box holds.

Level 1 support

Start here: Find each volume with $V = l \times w \times h$, then subtract the smaller from the larger.

The large box volume is ____ in^3 because $24 \times 18 \times 18 =$ ____.

El volumen de la caja grande es ____ in^3 porque $24 \times 18 \times 18 =$ ____.

The large box holds ____ more cubic inches because ____ - ____ = ____.

La caja grande contiene ____ pulgadas cúbicas más porque ____ - ____ = ____.

Word bank: volume cubic units rectangular prism length width height

Listen for: Students compute large = 7776 in^3 , medium = 3024 in^3 , then subtract to get 4752 in^3 more, and reason that you must find each volume before comparing.

Level 2

Why can't you decide which box holds more by comparing just one dimension (like length)? Use the two boxes to justify.

- Comparing only length is not enough because ____.
- Even though one box is longer in one edge, ____.

REFLECT

As you seal the capsule, a teammate writes a box's volume as 220 in^2 . How would you fix the units and explain why volume must use cubic units?

Level 1 support

Start here: Volume measures three-dimensional space, so the units must be cubic (in^3), not square (in^2).

The units should be ____ instead of in^2 because volume measures ____.

Las unidades deben ser ____ en lugar de in^2 porque el volumen mide ____.

Square units (in^2) are used for ____, but cubic units (in^3) are used for ____.

Las unidades cuadradas (in^2) se usan para ____, pero las unidades cúbicas (in^3) se usan para ____.

Word bank: volume cubic units rectangular prism edge net

Listen for: Students change in^2 to in^3 and explain that volume fills 3D space (three edges multiplied), while square units describe the flat area of one face.

Level 2

Why does multiplying inches $\times$ inches $\times$ inches produce cubic inches? Generalize the rule for any volume unit.

- Multiplying three lengths gives cubic units because ____.
- In general, the unit for volume is always ____ because ____.

Guided Examples

Example 1

What is the volume of a rectangular prism with $l = 7$ in, $w = 3$ in, $h = 4$ in?

Solution: $V = l \times w \times h = 7 \times 3 \times 4 = 84$ cubic inches.

Answer: A. 84 in^3

Example 2

A time capsule box has a volume of 120 cm^3 . Its length is 10 cm and width is 4 cm. What is its height?

Solution: $V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3 \text{ cm}$.

Answer: A. 3 cm

Example 3

Which unit is used for volume?

Solution: Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.

Answer: A. Cubic inches (in^3)

Write About the Math

The Writing Revolution

I can explain my work using the words volume, rectangular prism, cubic units, and edge.

1. Kernel Sentence subject + verb

Model: Volume is how much space is inside a solid shape.

Volumen es cuánto espacio hay dentro de una figura sólida.

Write a kernel sentence about volume. Use a subject and a verb.

Escribe una oración base sobre volumen. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Volume matters in math

Volumen importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Volume matters in math because ____.

Volumen importa en matemáticas porque ____.

but
pero

Volume matters in math, but ____.

Volumen importa en matemáticas, pero ____.

so
entonces

Volume matters in math, so ____.

Volumen importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about volume.

Di un hecho verdadero sobre volume.

Volume ____.

Question *Pregunta*

Ask a question about volume.

Haz una pregunta sobre volume.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about volume.

Muestra entusiasmo sobre volume.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with volume.

Dile a un compañero qué hacer con volume.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I need three measurements because volume fills space in ____ directions.

Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give one ____, not the whole volume.

El largo y el ancho solos solo dan una ____, no todo el volumen.

When the net folds up, the ____ faces enclose a ____ that I can fill with unit cubes.

Cuando la red se dobla, las ____ caras encierran un ____ que puedo llenar con cubos unitarios.

Try It

Solve on your own. Check the answer key when you are done.

1. Which unit is used for volume?

- A. Cubic inches (in^3)
- B. Square inches (in^2)
- C. Inches (in)
- D. Degrees ($^\circ$)

Show your work:

2. Two storage bins: Bin A is $8 \times 6 \times 5$ inches. Bin B is $9 \times 4 \times 7$ inches. Which holds more and by how much?

- A. Bin B by 12 in^3
- B. Bin A by 12 in^3
- C. Bin B by 24 in^3
- D. They hold the same

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Sentence starter: Option 1: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 2: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 3: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$.

The $\underline{\hspace{1cm}}$ option is closest to a cube because $\underline{\hspace{1cm}}$. This matters because $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 11$ in, $w = 5$ in, $h = 4$ in. What is the volume?

- A. 220 in^3
- B. 55 in^3
- C. 220 in^2
- D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Cubic inches (in^3) — *Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.*
2. **Try It 2:** A. Bin B by 12 in^3 — *Bin A: $8 \times 6 \times 5 = 240 \text{ in}^3$. Bin B: $9 \times 4 \times 7 = 252 \text{ in}^3$. Bin B holds 12 in^3 more.*
3. **Exit Ticket:** A. 220 in^3 — *$V = 11 \times 5 \times 4 = 220$ cubic inches. Remember: volume uses cubic units (in^3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Volume is how much space is inside a solid shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).